

Little Red Dots: Super-Eddington Winds, Not Overmassive Black Holes

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Type IIn supernovae and eta Carinae's Great Eruption offer a stellar template for JWST's enigmatic Little Red Dots.

Background

JWST's Little Red Dots show a unique combination of features: blackbody-like continua, broad Balmer lines, and no variability. No other galaxy or AGN class displays all of these simultaneously.

Are LRD broad lines virial, implying overmassive black holes, or do dense winds mimic a broad-line region?

What this paper is selling

The authors argue that dense, slow winds enshrouding a buried engine reproduce LRD spectra, paralleling Type IIn supernovae and eta Carinae's Great Eruption. This wind physics may inflate black hole mass estimates.

Key results to remember

- Escape velocity argument caps LRD engine masses below 10^5 solar masses.
- Pseudo-photospheres near 5000 K arise from hydrogen recombination self-regulation.
- Dust in expanding envelopes may seed classical AGN and massive black holes.

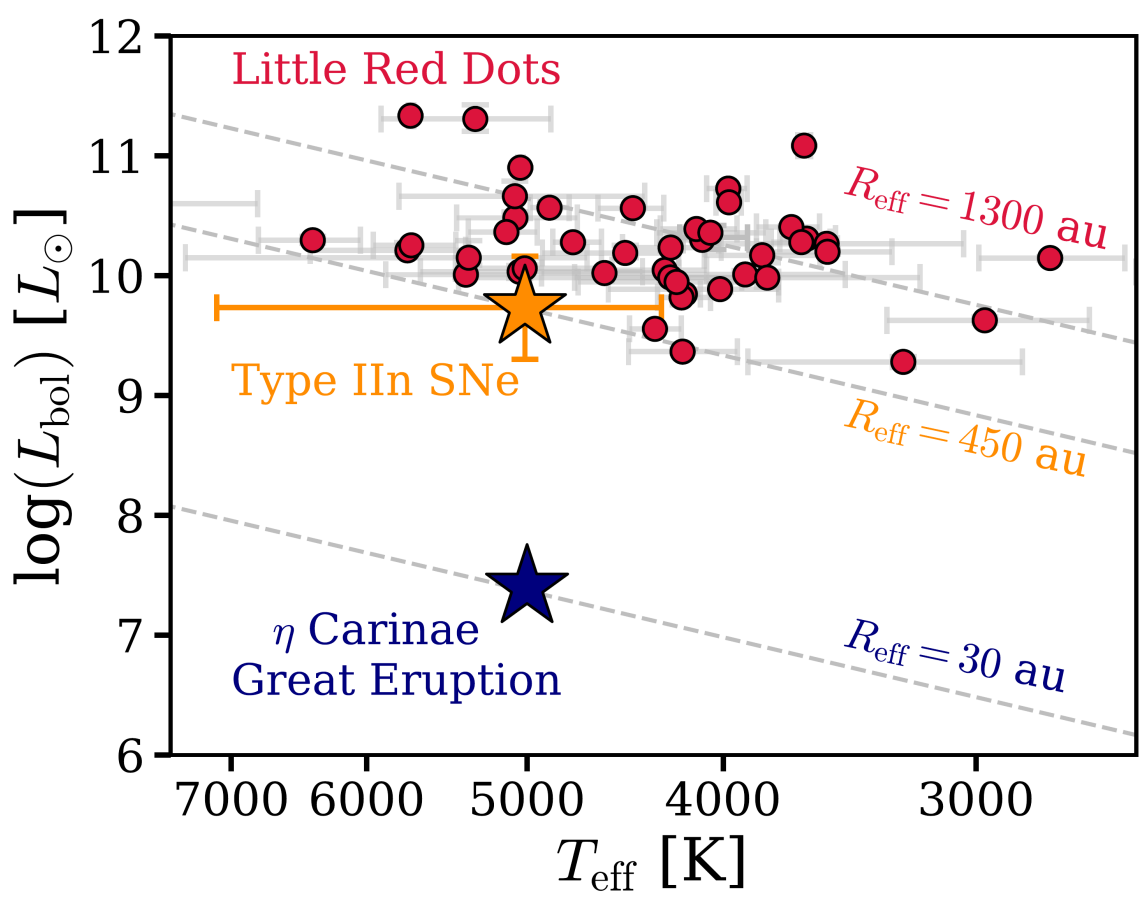


Figure 4 | LRDs, Type IIn supernovae, and eta Carinae cluster at similar effective temperatures despite spanning four decades in luminosity.

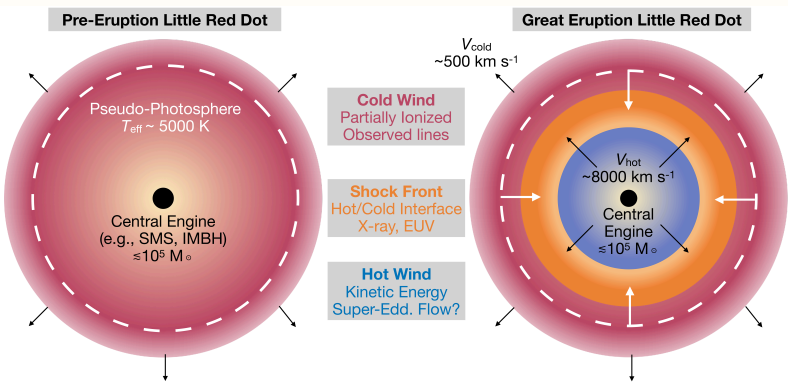


Figure 8 | Schematic showing a cold wind enshrouding the engine, then hot wind shocks converting kinetic energy into LRD radiation.